

Higher-degree arithmetic sieves

Rank-one completion, restoration, and exact cubic certificates

Arithmetic sieve supplement | 7 October 2026

This supplement proves an all-degree restoration interface and a conditional rank-one finite-state completion theorem, then gives exact cubic examples in $\mathbb{Z}[\theta]$, $\theta^3=2$. Their periodic avoiding-component maxima are 6 and 56 for two explicitly stated step sets, at supplied period-ideal indices 30 and 330. The proofs and accompanying certificates are self-contained apart from clearly identified standard arithmetic inputs: Dirichlet/S-unit ranks, ideal norms, Beukers-Schlickewei, ray class fields, and Chebotarev.

No arbitrary-step, 26-neighbor, universal higher-degree moat, or globally minimal-period theorem is asserted. Exact-integer computational checks and an independent mathematical audit accompany the proof. They are not a formal proof verification or an assessment of mathematical priority.

1. The useful new conditional engine

Let O be an order in a number field K of degree n , with its additive group identified with $\mathbb{Z}^n$. Let F be a finite symmetric set of nonzero coefficient steps. Suppose a finite list of nonzero nonunit principal generators gives a periodic avoiding set A_0 . Choose any supplied finite-index additive period lattice L , so $A_0+L=A_0$. The period need not be scalar.

Construct the **complete** finite quotient graph on A_0/L , retaining directed step labels, loops and parallel edges. For each connected quotient component C choose a spanning tree, quotient representatives r_v , and tree potentials t_v in L . Put $y_v=r_v+t_v$, with signs chosen so every tree edge lifts exactly. For every directed edge $e:v \rightarrow u$ with step f , put

$$h_e = y_v + f - y_u \text{ in } L.$$

Let H_C be the subgroup generated by all h_e . This is exactly the subgroup of closed-walk voltages: each non-tree edge closes with tree paths, whose voltages are zero, while every closed walk has voltage equal to the sum of its directed-edge voltages. Symmetry of F permits reversed walks, so subgroup generation and realizable closed-walk translations coincide. All these objects can be computed using exact integer linear algebra.

Theorem 1. Rank-one finite-state completion

Assume $\text{rank}(H_C) \leq 1$ for every quotient component C . Assume also the arithmetic supply of infinitely many distinct rational primes p admitting a principal ideal (α_p) of index p in O . Then finitely many additional such principal ideals make the F -step avoiding graph have only finite components.

Here are verifiable requirements and an explicit bound.

- If $H_C=0$, put $V_C=|C|$. Its lifted components already have V_C vertices.
- If $\text{rank}(H_C)=1$, compute a nonzero generator w_C of H_C and exact integers k_e with $h_e=k_e w_C$. Put $K_C=\max_e |k_e|$, which is at least 1.
- For each pair (v,j) with v in C and $0 \leq j < K_C$, select a different rational prime $p_{\{C,v,j\}} > K_C$ and a principal generator $\alpha_{\{C,v,j\}}$ of norm $\pm p$. Require $p_{\{C,v,j\}}$ not to divide $N(w_C)$. Use distinct rational primes within this component; choosing all primes distinct across components is allowed.
- Put $P_C=\text{product}_{\{v,j\}} p_{\{C,v,j\}}$.

For the enlarged sieve one may take

$$B = \max(\{1\} \cup \{V_C : H_C=0\} \cup \{V_C(P_C-K_C) : \text{rank}(H_C)=1\}).$$

All the finite-state hypotheses, arithmetic witnesses, norms, exclusions and bounds are exact checks. The prime-supply input guarantees these finite choices exist. It does **not** guarantee existence of a rank-reducing initial gate sieve.

Proof

For $H_C=0$, integrating all voltages gives finite lifted components, each with one point $y_v+\lambda$ per quotient vertex v .

For rank one, every lifted component above C consists exactly of

$$\{ y_v + \lambda + m w_C : v \in C, m \in \mathbb{Z} \},$$

for a suitable translate λ in L . This follows by lifting tree paths and then closed walks: the attainable period translations are precisely H_C . The representations are unique: different phases have different residues modulo L , and w_C is nonzero. Each edge changes the level m by one of the integers k_e , hence by at most K_C .

The ring $O/(\alpha_{\{C,v,j\}})$ has p elements and is therefore F_p . Write its surjective residue map as $\chi_{\{C,v,j\}}$. The exclusion p not dividing $N(w_C)$ ensures $\chi_{\{C,v,j\}}(w_C)$ is nonzero: otherwise $w_C=\alpha z$ would imply p divides $N(w_C)$, by multiplicativity of the field norm.

For this particular lifted component, require the entire K_C -level slab

$$\begin{aligned} & y_v + \lambda + (m_0+j) w_C, \\ & v \in C, \quad 0 \leq j < K_C, \end{aligned}$$

to be deleted. Assign its (v,j) point to the corresponding principal ideal. Each membership condition is the single solvable linear congruence

$$m_0 \chi(w_C) = -\chi(y_v+\lambda+j w_C) \pmod{p_{\{C,v,j\}}}.$$

The coefficient of m_0 is invertible. Distinct rational primes make the Chinese remainder theorem applicable, giving one m_0 modulo P_C satisfying all these conditions simultaneously. The chosen global sieve therefore deletes that full slab and every translate of it by $P_C w_C$. The starting congruence can depend on λ ; the list of excluded principal ideals does not.

A deleted slab of K_C consecutive levels blocks every edge from the levels below it to the levels above it: such an edge would have level change at least K_C+1 . Between consecutive slabs there are P_C-K_C levels and V_C vertices per level. Every surviving connected component has at most $V_C(P_C-K_C)$ vertices. Other deleted points only reduce components. This proves the theorem.

Arithmetic supply in every order

Here is the precise class-field input and its application. Choose a positive integer f with $f O_K$ contained in O . Let M be the ray class field of K for the finite modulus $f O_K$, and let E/Q be a finite normal closure containing M . The class-field existence and Artin reciprocity theorems identify the ray class group with $\text{Gal}(M/K)$. If an unramified rational prime p splits completely in E , each degree-one prime ideal $\mathfrak{p}_K$ of O_K above p has trivial Artin symbol in M/K . Its ray class is therefore trivial, so $\mathfrak{p}_K=(\alpha)$ for an integral α with α congruent to 1 modulo $f O_K$. Consequently α belongs to O and $|N_{\{K/Q\}}(\alpha)|=p$.

The multiplication lattice αO has index p : its determinant in any $\mathbb{Z}$ -basis of O is the field norm. Its inclusion in the kernel of the residue map $O \rightarrow O_K/\mathfrak{p}_K$ is automatic. The latter map is onto F_p because it contains the residues of rational integers, so its kernel also has index p . The two lattices are equal. Thus αO is a principal index- p ideal in O , even when O is nonmaximal. Chebotarev applied to E/Q supplies infinitely many distinct such rational primes. Removing the finitely many ramified primes and primes dividing f does not change this conclusion.

The deep imported inputs are ray-class-field existence/Artin reciprocity and Chebotarev, rather than an effective search bound. See J. S. Milne, [Class Field Theory, V.3.6 and V.3.23](#). No class-number-one assumption is being made.

Exact algorithmic scope

The ray-class/Chebotarev principal split-prime supply proved from the standard imported arithmetic inputs below provides the required infinite set for every number-field order, including nonmaximal orders. Avoiding finitely many primes dividing $N(w_C)$, and choosing distinct primes larger than K_C , preserves that supply. The present graph proof uses no further analytic input.

Equivalently, Theorem 1 can be used without any existence theorem by supplying and checking the displayed finite generators directly. Given a certified rank-one gate, enumeration of integral coefficient vectors, exact norm-prime tests and the finite exclusions gives a terminating completion search under that imported supply. This is not a terminating algorithm for arbitrary F : no theorem here constructs the initial rank-at-most-one gate for arbitrary F .

The rank test uses closed-walk voltages, not the rank of the quotient vertex set, apparent narrowness of a picture, or finiteness of the quotient. The spanning-tree shifts and the gcd needed to obtain a generator of H_C must be included; a merely primitive vector on the same rational line can be too small.

2. Explicit cubic arithmetic

Let

$$0 = Z[\theta], \theta^3=2,$$

with coefficient basis $(1, \theta, \theta^2)$. The polynomial is irreducible by Eisenstein at 2, so this is a degree-three field order. Multiplication by $\alpha = a + b\theta + c\theta^2$ has integer matrix

$$\begin{bmatrix} a & 2c & 2b \\ b & a & 2c \\ c & b & a \end{bmatrix},$$

and determinant

$$N(\alpha) = a^3 + 2b^3 + 4c^3 - 6abc.$$

Take

$$\begin{array}{ll} \alpha_2 = \theta, & N=2, \\ \alpha_3 = 1 + \theta, & N=3, \\ \alpha_5 = 1 + \theta^2, & N=5. \end{array}$$

Their ideals have the respective exact kernels

$$\begin{array}{l} a \equiv 0 \pmod{2}, \\ a - b + c \equiv 0 \pmod{3}, \\ a + 3b + 4c \equiv 0 \pmod{5}. \end{array}$$

To prove each equality, multiplication of every matrix column gives residue zero, the residue map is surjective (the coefficient of a is 1), and the matrix determinant has absolute value p . Thus image and kernel have the same index p . No ideal factorization or class-number assertion is needed.

Their product generator is

$$\begin{array}{l} \gamma_{30} = \theta(1 + \theta)(1 + \theta^2) = 2 + 3\theta + \theta^2, \\ N(\gamma_{30}) = 30. \end{array}$$

The ring map

$$\phi_{30}(a, b, c) = a + 8b + 4c \pmod{30}$$

is well-defined since $8^3 \equiv 2 \pmod{30}$ and $8^2 \equiv 4 \pmod{30}$. It is surjective. Its kernel is $\gamma_{30} \mathcal{O}$: inclusion follows by multiplying matrix columns, and both lattices have index 30. Reduction modulo 2, 3, 5 gives exactly the three kernels above. The avoiding quotient therefore consists of the eight units modulo 30. The supplied period ideal has index 30 and contains $30 \mathcal{O}$, so 30 is a common scalar period for this fixed sieve. No period optimality across other sieves or choices of generators is asserted.

3. Six-axis-step cubic certificate: exact $B=6$

Define the meaningful full-rank coefficient step set

$$F_6 = \{+/-1, +/-\theta, +/-\theta^2\}.$$

The complete avoiding quotient modulo $\gamma_{30} O$ has precisely two path components:

```
13 -- 17,
29 -- 7 -- 11 -- 19 -- 23 -- 1.
```

All steps ± 1 are forbidden by parity. The other step residues are ± 8 and ± 4 . Exact lift sections for the eight states are

```
1: (1, 0, 0)
7: (1, -2, -2)
11: (1, -2, -1)
13: (1, 1, 1)
17: (1, 1, 2)
19: (1, -1, -1)
23: (1, -1, 0)
29: (1, -3, -2).
```

Every allowed directed edge $r \rightarrow s$ with step f satisfies $y_r + f = y_s$ as an identity in $\mathbb{Z}^3$. Thus all cycle voltages vanish, and every lifted avoiding component is a translate by $\gamma_{30} O$ of one of the two finite displayed sections. The exact maximum component cardinality is $B=6$.

The dimension-reduction proof, without a search

Before excluding α_5 , parity fixes the odd constant coefficient a . Put $T=a+3k+1$. Avoiding α_3 restricts $b-c$ to the two adjacent values $T, T+1$ for some integer k . Since an axis step changes $b-c$ by ± 1 and cannot cross the forbidden congruence class, every gate component is a doubly infinite zigzag:

```
L_m = (a, m+T, m),
U_m = (a, m+T+1, m),
... L_m -- U_m -- L_{m+1} ...
```

It has translation direction $w = \theta + \theta^2$, of norm 6. Modulo α_5 ,

```
chi_5(L_m) = a+3T+2m,
chi_5(U_m) = a+3T+2m+3.
```

If L_l is forbidden, then U_{l+1} is forbidden. All forbidden positions in the path index $j=2m$ for L_m and $j=2m+1$ for U_m are

```
j=2l or 2l+3 mod 10.
```

The cyclic gaps are 3 and 7, so the surviving path runs have sizes 2 and 6. This proves the certificate structurally and explains why no scalar-grid period tuning was needed.

4. A sharp obstruction to one-generator completion

Enlarge the steps to

```
F8chain = F6 union {+/- (theta + theta^2)}.
```

This is an explicitly defined eight-step full-rank coefficient graph. It is not the 26-neighbor coefficient graph.

Under the first two gate ideals (θ) and $(\theta+1)$, each zigzag becomes the square of a bi-infinite path: every path-distance 1 edge is an axis step, and every path-distance 2 edge is $\pm w$. The following is stronger than failure of one tested period.

Proposition 2. One additional principal ideal never suffices

For this fixed gate and $F8chain$, adjoining any single nonzero nonunit principal ideal (α) leaves an infinite avoiding component.

Proof. Consecutive zigzag vertices differ by θ or θ^2 . If both belong to (α) , then α divides θ or θ^2 . The element θ is prime since $O/(\theta) = \mathbb{F}_2$. A nonunit divisor of θ^2 must be divisible by θ : if $\alpha \theta = \theta^2$ and θ does not divide α , primality forces $\theta = \alpha \theta_1$, then $\alpha \theta_1 = \theta$, then

$\beta_1 = \theta \beta_2$, making α a unit. The same reasoning handles a divisor of θ . Hence (α) is contained in (θ) , contradicting the fact that both gate vertices have odd constant coefficient.

Thus the vertices removed by one additional ideal are never consecutive on the underlying path. Each missing vertex is bypassed by the path-distance2 edge, so all remaining path vertices are connected and infinitely many remain. This proves the claim.

In particular the norm5 ideal alone cannot finish F8chain. An explicit 8-step nonzero-period witness for the norm2,3,5 sieve is

$(1, -1, 0), (1, 0, 0), (1, 1, 1), (1, 1, 2), (1, 2, 3),$
 $(1, 3, 3), (1, 3, 4), (1, 4, 4), (1, 4, 5).$

Its displacement is $5w = (0, 5, 5)$, which is in $\gamma_{30} O$. Every point avoids all three ideals. Periodic concatenation gives an infinite avoiding walk. This says nothing about existence of an infinite walk on irreducible elements.

5. Two-prime CRT slabs finish the eight-step cubic graph

Add

$\alpha_{11} = -1 + \theta + \theta^2, N(\alpha_{11}) = 11.$

Its kernel is $\theta \rightarrow 7 \pmod{11}$, namely $a + 7b + 5c = 0 \pmod{11}$. This follows by the same matrix-index proof; $7^3 = 2 \pmod{11}$ and $7^2 = 5 \pmod{11}$.

The gate components above have two phases L_m, U_m and level jumps of size at most 1. To delete an entire level, require

$L_m \text{ in } (\alpha_5), \quad U_m \text{ in } (\alpha_{11}).$

The conditions are

$m = -3(a + 3T) \pmod{5},$
 $m = -a - 7T - 7 \pmod{11}.$

CRT gives one such completely deleted level modulo 55 for every a, T . These levels block all F8chain edges. The structural bound is at most $2(55-1) = 108$. This is an explicit instance of Theorem1 with $V=2$ and $K=1$.

The independent exact quotient certificate sharpens 108 to 56. The four-ideal product is

$\gamma_{330} = \gamma_{30} \alpha_{11} = 6 + \theta + 4\theta^2,$
 $N(\gamma_{330}) = 330.$

Use the surjective ring map

$\phi_{330}(a, b, c) = a + 128b + 214c \pmod{330},$

where $128^3 = 2$ and $128^2 = 214$ modulo 330. Its kernel is $\gamma_{330} O$ by the exact matrix-index proof. The complete allowed quotient is the 80 units modulo 330. The eight-step residues are $\pm 1, \pm 128, \pm 214, \pm 12$. There are 216 allowed directed edges and four quotient components, of sizes

6, 9, 9, 56.

Unlike the F6 quotient, this graph has 32 independent undirected cycles. Consequently merely listing finite quotient components is insufficient. The certificate supplies an actual integer lift section for every one of its 80 states. Every one of its 216 directed edges satisfies $y_r + f = y_s$ exactly. All cycle voltages therefore vanish and the exact avoiding component bound is $B=56$. The section identities are independently checked, not inferred from a finite exploration cutoff.

6. A stronger associate-class restoration interface

A norm-cutoff restoration theorem is given below. The stronger interface in this section uses the fact that only finitely many associate classes are exceptional, together with the exact Beukers-Schlickewei theorem. It avoids the S-unit rank contributed by primes in the generator norms.

Theorem 3. Uniform intersections of finitely many associate classes

Let $U=O^\times$ have multiplicative rank r , let $q_1, \dots, q_k$ be nonzero elements of O , and put $E=\bigcup_i q_i U$. For every nonzero v in K ,

$$|\{x \in E : x+v \in E\}| \leq k^2 \cdot 2^{16(r+1)}.$$

The bound is independent of the sizes, norms and prime divisors of v and q_i . It remains an upper bound for every subset of E . Overlap between associate classes only overcounts the total.

For every order of signature (r_1, r_2) , $r=r_1+r_2-1$. To justify this beyond maximal orders, choose f with $f \in O_K$ contained in O . The kernel of the finite reduction map $O_K^\times \rightarrow (O_K/f O_K)^\times$ has finite index. A unit in this kernel and its inverse are both congruent to 1 modulo $f O_K$ and belong to O , so this kernel lies in $O^\times$. Thus $O^\times$ has the same rank as $O_K^\times$. The latter rank is Dirichlet's unit theorem. This argument does not assume that elements outside O with norm ± 1 are units of O .

Proof. Fix an embedding of K in $\mathbb{C}$. For each ordered pair i, j , a solution $x=q_i u$ and $x+v=q_j z$, with u, z in U , gives

$$(X, Y) = (-q_i u/v, q_j z/v), \quad X+Y=1.$$

These pairs lie in the subgroup of $(\mathbb{C}^\times)^2$ generated by $U \times U$ and the single pair $(-q_i/v, q_j/v)$. Its rank is at most $2r+1$. The exact Theorem 1.1 of Beukers-Schlickewei bounds solutions in its Q -closure by $2^{8(2r+2)}$, and the subgroup itself is contained in that closure. The map from x to (X, Y) is injective. Sum the bound over the k^2 ordered pairs. Torsion contributes no rank; the normalization coefficients do not require additional places.

The imported primary theorem has the hypothesis of the Q -closure of a finitely generated rank- R subgroup of $(\mathbb{C}^\times)^2$, and its numerical bound is 2^{8R+8} . [Primary paper, Theorem 1.1](#). The deep theorem is imported, not reproved.

Corollary 4. Norm-independent restoration from k generators

Suppose a finite list of k nonzero nonunit principal generators gives an F -step avoiding graph whose components have at most B vertices. Put

$$\begin{aligned} D &= \max\{|f| : f \in F\}, \quad \Delta = |F|, \\ L_n(R) &= (2R+1)^{n-1}, \\ C &= k^2 \cdot 2^{16(r+1)}. \end{aligned}$$

Then every irreducible F -step component has at most

$$[1 + L_n(D(B+1)) \cdot C] \cdot [1 + \Delta \cdot B]$$

vertices. The empty-generator and empty-step cases retain their obvious stronger bounds B and 1 respectively.

Proof. An irreducible divisible by a supplied nonunit q_i is q_i times a unit, so the exception set lies in E . For each of the $L_n(R)$ nonzero coefficient differences with sup-norm at most R , Theorem 3 bounds its ordered exception edges by C . Therefore at most $L_n(R)C$ exception vertices are nonisolated; each exception component has at most $1 + L_n(R)C$ vertices. Take $R=D(B+1)$. Two exception vertices touching the same ordinary component are within R : that component has at most B vertices, so a shortest connecting ordinary path has at most $B-1$ edges, plus the two exception edges. Every exception vertex in one full component is therefore in the same enlarged exception component. Each such vertex touches at most Δ ordinary components, each of cardinality at most B . This gives the displayed bound, exactly as in the preceding restoration proof.

For this cubic signature $(1, 1)$, $r=1$. Our three-generator F_6 certificate gives

$$37(1+30366 \cdot 2^{32}) = 4,825,576,145,682,469.$$

Our four-generator F_8 chain certificate gives

$$449(1+24333984 \cdot 2^{32}) = 46,926,635,792,162,881,985.$$

These remain very conservative irreducible cardinality bounds. The smaller numbers $B=6$ and $B=56$ apply to the periodic avoiding graphs. No practical moat radius, optimal irreducible bound or all-finite-step conclusion is claimed.

The broader bounded-norm theorem remains useful separately: a bounded-norm set need not be supplied as a small fixed list of associate classes. The new corollary is an improved interface for the specific exception set arising from a finite principal sieve.

7. Bounded-norm intersections and arbitrary-order restoration

This complementary result applies when an exceptional set is not presented as a short fixed list of associate classes. Let K have degree n and signature (r_1, r_2) , and let O be any order. For an integer $M \geq 1$ put

$$E_M = \{x \in O : 0 < |N_{K/Q}(x)| \leq M\}.$$

Let S_M be the finite prime ideals of O_K above rational primes at most M , and let U_M be the S_M -unit group. Its rank is

$$r_M = r_1 + r_2 - 1 + |S_M|.$$

Theorem 5. Translation-uniform bounded-norm intersections

For every nonzero v in K ,

$$|\{x \in E_M : x+v \in E_M\}| \leq C_M = 2^{\{16(r_M+1)\}}.$$

One may use the coarser field-uniform bound

$$C(n, M) = 2^{\{16 n (\pi(M)+1)\}} \geq C_M.$$

Here $\pi(M)$ counts rational primes at most M . The constants are uniform in v and in the chosen order and coefficient basis for fixed K and M ; the coarser constant is uniform over fields of fixed degree n . They are not independent of both degree and M . The condition $v \neq 0$ is essential.

Proof. For x in E_M the principal integral ideal $x O_K$ has numerical norm $|N(x)|$. Every finite prime ideal dividing it lies over a rational prime at most M . Consequently x is an S_M -unit; the same holds for $y=x+v$ in E_M . This takes place in O_K and uses no ideal factorization in the possibly nonmaximal order O .

For one embedding $K \rightarrow \mathbb{C}$, the pair $(-x/v, y/v)$ belongs to the subgroup generated by $U_M \times U_M$ and $(-1/v, 1/v)$. The product group has rank $2r_M$ and the one added pair raises rank by at most one. It satisfies $X+Y=1$ and determines x injectively. Beukers-Schlickewei therefore gives $2^{\{8((2r_M+1)+1)\}} = 2^{\{16(r_M+1)\}}$. No places dividing v were added. Since at most n prime ideals lie above each rational prime and $r_1 + r_2 \leq n$, we have $r_M + 1 \leq n(\pi(M)+1)$, proving the coarser bound.

For the standard ideal-norm and unit/ S -unit rank inputs see J. S. Milne, [Algebraic Number Theory, Proposition 4.2, Theorems 5.1 and 5.11](#). The Beukers-Schlickewei theorem is imported with the exact scope stated in Section 6; its proof is not claimed here.

Lemma 6. Exceptional component counting

Suppose E is any set with at most C points x such that $x, x+v$ belong to E , for every nonzero coefficient difference v . Join points of E when their coefficient sup-norm distance is at most an integer $R \geq 0$. There are $L_n(R) = (2R+1)^{n-1}$ possible nonzero ordered differences. Thus the graph has at most $L_n(R)C$ ordered edges. Every nonisolated vertex is the initial endpoint of such an edge, so at most $L_n(R)C$ vertices are nonisolated. Each component has at most $1 + L_n(R)C$ vertices. This harmless extra one also covers isolated vertices. The statement holds for every subset of E .

Theorem 7. General restoration interface

Let F be a nonempty finite symmetric set of nonzero coefficient steps, $D = \max_{f \in F} \|f\|_{\infty}$ and $\Delta = |F|$. Let $\alpha_1, \dots, \alpha_k$ be nonzero nonunits of O . Suppose the F -step avoiding graph on $A = O \setminus \bigcup_i \alpha_i O$ has components of cardinality at most $B \geq 1$. An irreducible here means a nonzero nonunit x such that $x=ab$ forces one of a, b to be a unit in the domain O .

If the exceptional irreducibles are contained in a set E with the intersection bound C of Lemma 6, every irreducible F -step component has at most

$$\max\{B, [1+L_n(D(B+1))*C] [1+\Delta B]\}$$

vertices. For the principal-generator sieve one may take either

$$C = k^2 \cdot 2^{\{16(r+1)\}}$$

or

$$C = C_M, \quad M = \max_i |N(\alpha_i)|.$$

An empty generator list gives the stronger bound B directly; an empty step set gives cardinality 1 for every component.

Proof. If irreducible $x = \alpha_i u$, nonunit α_i forces u to be a unit of O . Thus exceptional irreducibles lie in $\bigcup_i \alpha_i O^*$ and in E_M , since $N(u) = \pm 1$. Every other irreducible lies in A ; its induced graph has components of at most B vertices and degree at most Δ .

Enlarge the exception graph using coefficient radius $R = D(B+1)$. If two exceptional vertices touch a common ordinary component, its two adjacent ordinary vertices can be joined by a path of at most $B-1$ edges. Including the two incident exception edges gives at most $B+1$ steps, hence distance at most R . Consecutive exceptions along any finite original path are therefore adjacent in this enlarged graph, either directly or through one ordinary component. Consequently all exceptions in one original component lie in one enlarged exceptional component and there are at most $M_E = 1 + L_n(R)C$ of them.

Each exception touches at most Δ ordinary components, each with at most B vertices. Counting these ordinary vertices and the exceptions gives $M_E(1 + \Delta B)$. A component without exceptions has at most B vertices. This argument uses finite paths within a connected component and does not assume in advance that the component is finite. It proves the stated cardinality bound, including exclusion of infinite components.

What a periodic certificate must prove

For nonzero α , multiplication on the coefficient lattice has determinant $N(\alpha) \neq 0$, so αO has finite index and contains some scalar sublattice $Q \subset O$. A finite family therefore has a common finite-index additive period. This ensures a finite quotient is available, but its finiteness is not itself a bound for the lifted graph. Every directed allowed quotient edge must carry its step label and exact voltage. A section with zero voltage on every edge proves finite lifted components of the exact quotient sizes. Conversely a closed walk with nonzero voltage lifts to arbitrarily large translations and an infinite component. This criterion holds in every lattice rank; it does not assert existence of a successful principal sieve for every F .

8. What has and has not been established

Established:

1. A complete rank-at-most-one periodic voltage gate can be completed using finitely many norm-prime principal ideals, with an explicit CRT slab bound.
2. A full-rank degree-three six-axis-step principal sieve with exact $B=6$.
3. A full-rank degree-three eight-step principal sieve with exact $B=56$, obtained by a finite-state mechanism beyond degree-two gates.
4. An impossibility theorem for every single additional principal generator in the fixed two-ideal eight-step gate, explaining why multiple-prime CRT slabs are needed.
5. A norm-independent finite-associate-class restoration interface, with explicit improved irreducible bounds for both examples.

Not established:

- An initial rank-one gate for every finite F in a cubic field or order.
- The 26-neighbor graph, arbitrary bounded coefficient steps, a universal higher-degree moat theorem, or a general higher-arity entropy enrichment.
- Optimal scalar periods, independent mathematical priority, or formal proof verification. The examples are confined to the displayed coefficient basis.

9. Certificate contract and reproducibility

The package contains the unchanged complete JSON certificates `cubic_f6_certificate.json` and `cubic_rank_one_f8_certificate.json`. Every allowed state has one exact integer lift section; steps and quotient data are explicit. The public author-derived checkers reconstruct all allowed directed step-labelled edges and verify every section identity. Their metadata schema now rejects booleans and floating-point replacements of exact integers. The original verifier bytes are preserved separately, and exact unified diffs document the changes.

`independent_check.py` imports no author verifier or search code. It uses polynomial reduction for multiplication, the Leibniz determinant formula, and Gaussian elimination over exact Fractions for ideal membership. It reconstructs the quotient graphs and the complete two-ideal F8chain voltage gate. Its 15 targeted negative controls per certificate include wrong arithmetic, omitted or duplicate states, changed steps, and a same-residue section change with nonzero voltage. It also strictly rejects boolean/float metadata.

The 729 vectors in $[-4,4]^3$ provide 2,916 F6 and 3,645 F8 ideal-membership cross-checks. These finite checks are auxiliary controls: global kernel equalities follow from exact inclusion and lattice index, and lifted component bounds follow from all-edge section identities. No finite exploration cutoff is used as proof of global boundedness.

Run `python3 tools/reproduce.py` from the package directory. It verifies the exact whitelist, hashes, derivative diffs, both checkers and their independent counterpart normally and with `python3 -O`, and the saved result bytes. It can also create a deterministic tar.gz archive. The standard-library Python checks need no external packages. `README.md` gives commands and source-map details.

The executable checks are reproducibility aids and exact finite witnesses; the mathematical argument is the lifting, lattice-index, arithmetic, and restoration proof above.